

PROFESSIONAL SUMMARY

Computer Science graduate from Ton Duc Thang University focused on machine learning, deep learning, recommendation systems, and distributed data processing. Pragmatic engineer who values inspecting intermediate model representations (attention weights, convolutional feature maps, Spark partitions) over black-box usage. Experienced in building testable software pipelines, multimodal architectures, and instructing programming fundamentals at MindX.

EDUCATION

Ton Duc Thang University / Bachelor of Science in Computer Science 2022 – 2026 · HCMC, Vietnam
Cumulative GPA: 3.33 / 4.00 · Focus: Artificial Intelligence, Machine Learning & Distributed Systems

TECHNICAL COMPETENCIES

CORE PROGRAMMING:	Python, SQL, C++, Java, JavaScript
DATA & ANALYSIS:	Pandas, NumPy, EDA, Feature Engineering, Data Preprocessing, Statistical Analysis
MACHINE LEARNING:	Scikit-learn, Supervised & Unsupervised Learning, Classification, Clustering, Recommendation Systems
DEEP LEARNING:	PyTorch, TensorFlow / Keras, CNN, RNN, LSTM, Seq2Seq Architectures, Attention Mechanisms
APPLIED AI & VISION:	Computer Vision (YOLOv8, ResNet), NLP (BERT), Multimodal AI, LLM Integration (Gemini API)
BIG DATA:	Apache Spark (PySpark), Distributed RDD Operations, MapReduce Concepts, Distributed Data Mining
TOOLS & METHODS:	Git, GitHub, Docker, Linux / Bash, Jupyter Notebook, SQL Server, OOP, Agile / Scrum

SELECTED TECHNICAL PROJECTS

Visual Question Answering on Vehicles / AI Engineer github.com/TranNhatMinh2012/deeplearning-midterm
Stack: PyTorch · YOLOv8 · ResNet · BERT · LSTM · Cross-Modal Attention · Seq2Seq

- Engineered an automated multimodal dataset pipeline combining YOLOv8 bounding proposals with Gemini API prompting for QA synthesis.
- Extracted 2048-dimensional spatial feature maps with ResNet backbones and modeled linguistic queries via Bi-LSTM and BERT embeddings.
- Implemented custom PyTorch Seq2Seq decoder modules with cross-modal attention dynamically aligning visual regions with query tokens.

Smart Retail Recommendation System with LLM Integration / Data & ML Engineer github.com/TranNhatMinh2012/DACNTT
Stack: Python · Pandas · Scikit-learn · TF-IDF · Co-occurrence Matrix · Gemini API

- Analyzed multi-item customer transaction baskets to identify primary driver products initiating shopping missions.
- Constructed statistical co-occurrence correlation matrices and catalogue TF-IDF similarities to rank high-affinity follower items.
- Integrated Gemini API for contextual re-ranking, balancing catalogue novelty, diversity, and basket synergy with natural-language rationales.

Distributed Data Mining with PySpark / Big Data Engineer github.com/TranNhatMinh2012/mining-massive-datasets-endterm
Stack: Apache Spark (PySpark) · K-Shingling · Jaccard Distance · Min-Heap · Agglomerative Clustering

- Built a distributed text mining pipeline decomposing unstructured documents into k-shingles and computing pairwise Jaccard distances.
- Engineered a min-heap priority queue to track nearest merge pairs in O(1) time complexity without executor memory overflow.
- Implemented hierarchical agglomerative clustering across distributed Spark RDD partitions utilizing broadcast variables.

Vietnamese–English Neural Machine Translation / Deep Learning Engineer github.com/TranNhatMinh2012/natural-language-processing
Stack: PyTorch · Seq2Seq · Bidirectional LSTM · Global Attention Mechanism

- Developed a sequence-to-sequence neural machine translation architecture with bidirectional LSTM encoder and attention decoder.
- Implemented Luong global attention alignment to compute dynamic attention weight distributions between source and target vocabularies.

• WORK EXPERIENCE

MindX Technology School / Lecturer, Part-time

Nov 2025 – Present • Ho Chi Minh City

- Instructed students in programming fundamentals, computational thinking, and algorithmic problem-solving using Python and C++.
- Mentored learners through structured debugging, clean code practices, project-based assignments, and Git version control workflows.